A Shorter Proof Method for Subset Sums

Endpoint transport, a coarse proof spine, and five exceptions

Sum-of-Subset research project

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Abstract

We simplify the independently audited proof that a finite real set A satisfying $\mathrm{FS}(A) \subseteq 4A$ has at most 97 elements. A monotone change of the two mass variables replaces every positive low-hole parameter by the endpoint 14. A coarse sumset estimate then handles 1,313 of 1,374 structural roots directly. Only four roots need the large-loss split; after refinement, only five of the 1,442 leaves need nested envelopes. The finite proof shrinks from 45,928 to 18,280 integer models, from 3,750 to 1,455 interval checks, and from 209 to 94 integer certificates. This is a simplification of the proof of $N \le 97$, not a new numerical upper bound. The independently closed range remains $71 \le N \le 97$; the stronger archived claim $N \le 84$ still requires its separate complete audit.

1 Scope and inherited structural reduction

For a finite real set A, define $\mathrm{FS}(A) = \{\sum_{x\in X} x : X \subseteq A\}$, including the empty sum. Repeated summands are allowed in $4A = A + A + A + A$. The previous report [1] excludes size 98 using the audited upper bound 98 and structural lemmas from the preceding chain. Here we retain its lattice proof, scalar hole bounds, covering units, and actual-set-to-model derivation, but replace the final linear proof. No previous endpoint certificate file is used by the new checker.

For a hypothetical size-98 set, the inherited reduction gives an integral 90-element set D, a core $E = P \cup \{0\} \cup (-Q)$ of size 82, and

$$a = |P|, \quad b = |Q| = 81 - a, \quad I = (a-40)(a-41), \quad \Omega = 395 - I, \quad T_r = r(r+1)/2.$$

Put $R_p = \max P$, $R_q = \max Q$, $M_p = \max D$, $M_q = -\min D$, and let S_p, S_q be the positive and negative-magnitude masses of D. The support of E has z holes; the support of D has $w \le 4z$ holes. The covering slack is $c = 2|2D| + 18 \cdot 90 + 85 - |\,\mathrm{FS}(D)|$. If j covering units are empty, set $\delta = c - 90j$. For a retained center v, let l_v, u_v be its lower and upper hull losses and $\lambda_v = l_v + u_v$. Then

$$\sum_{v \text{ retained}} \lambda_v \le \delta, \qquad 0 \le \lambda_v \le 89. \tag{1}$$

The common-lattice proof in [1] is essential: its refined inner comparisons are at most -388 for $z = 0$ and -121 for $z > 0$, whereas the discarded coarse comparison can equal 7. We do not replace that argument by a finite solver result.

At low holes, the inherited scalar bounds give

$$z = 0 : c \le 227 - I, \quad \varepsilon_p, \varepsilon_q \ge 0,$$
$$1 \le z \le 14 : c \le 174 - I + 2z, \quad \varepsilon_p \ge a - 3 + z, \quad \varepsilon_q \ge b - 3 + z,$$

where $\varepsilon_p = \sum P - T_a$, and similarly on the negative side. Thus $j \le 2$. For high holes the unchanged exact scalar audit excludes all-active states by a margin of at least 35. Only $(a, z) = (40, 56), (40, 57), (41, 56), (41, 57)$ allow an empty unit. They belong to H1, with

$$\delta \le 1, \quad \varepsilon_p \ge 240 + 9(a-40), \quad \varepsilon_q \ge 249 - 9(a-40), \quad S_p + S_q + \delta - 2K \le 1842, \tag{2}$$

where $K = |2D|$. There are 40 lattice orientations, 416 surviving low-hole states, and 1,574 surviving high-hole states, exactly as before.

2 Eliminate the hole variable by mass transport

The preceding proof retained the same real variable z in every positive low-hole row. It is possible to eliminate it without replacing correlated inequalities by unrelated worst-case bounds.

Lemma 2.1 (Monotone endpoint transport)**.** *Every feasible positive low-hole model with* $1 \le z \le 14$ *maps to a feasible model with* $z = 14$ *by the change*

$$t = 14 - z, \qquad \widehat{S}_p = S_p + t, \quad \widehat{S}_q = S_q + t,$$

with all other coordinates unchanged. This holds for global, core, and ordinary quantiles, and for every allowed large-loss refinement.

Proof. Here $t \ge 0$, so nonnegativity is preserved. The rows involving z or a mass variable have the following changes in their left sides:

Row or expression	Change
$\delta - 2z$	$-2t$
$R_p + R_q + \delta - 4z$	$-4t$
$2M_s + 6R_s - 2S_s + 2z$	0
$\sum_{i=1}^{4} x_i^s - S_s + z$	0
$M_s + 4R_s - S_s$, $\sum_{i=0}^{4} x_i^s - S_s$	$-t$
Upper quantile: $v - u_v - S_s$ plus unchanged terms	$-t$
$S_p + S_q + \delta - 2K - 4z$	$-2t$

All other rows are unchanged, except the hole bounds: the new value 14 satisfies both. Thus no upper-bound row is violated. The argument is uniform in a, the center choices, and the loss classification. □

The transported masses are auxiliary coordinates; they need not be masses of another actual set. This is sufficient for a necessary linear relaxation. The implication is from every actual low-hole state to the endpoint model, not an assertion that the actual hole count equals 14. In the rest of this report S_p, S_q denote these transported coordinates for positive low-hole models. Their mass bonuses are now

$$(\varepsilon_p, \varepsilon_q) \ge (a + 11, b + 11). \tag{3}$$

The verifier checks the transport direction in every source row before eliminating z. This algebraic check covers all real $t \ge 0$; it is not a sampling check at selected hole counts.

3 A coarse proof spine with no inner-point variables

The outer magnitudes are $p_1 > p_2 > p_3 > p_4 > M_p$ and their q reflection. The eight outer gaps are at least one when $j \le 1$; for $j = 2$ only weak order is used. Always $M_s \ge R_s + 4$ and $R_p \ge a$, $R_q \ge b$. Use the two mass lower bounds on each side:

$$S_p \ge T_a + \varepsilon_p + 3R_p + 6 + M_p, \qquad S_p \ge T_{a-1} + 4R_p + 6 + M_p. \tag{4}$$

Their reflections hold with p, a replaced by q, b. The bonuses are zero for Z, (3) for P, and (2) for H1. As in the previous verifier, the first mass row is multiplied by two and labelled H; the second is labelled Hhat.

Lemma 3.1 (Coarse double-sumset bound)**.** *For the integral set D,*

$$K \le M_p + M_q + R_p + R_q + 21.$$

Proof. Every pair except a pair of two positive outliers or two negative outliers lies in $[-M_q - R_q, M_p + R_p]$. This integer interval contains $M_p + M_q + R_p + R_q + 1$ positions. There are at most ten unordered repeated pairs among the four positive outliers and ten among the negative ones. Adding these possible exceptions proves the bound. □

The cardinality identity and $w \le 4z$ give $S_p + S_q + \delta - 2K \le 1704 - 90j + 4z$ at low holes. Add twice Lemma 3.1 to eliminate K entirely:

$$\boxed{S_p + S_q + \delta - 2(M_p + M_q + R_p + R_q) \le B,} \tag{5}$$

where $B = 1746 - 90j$ for Z, $B = 1802 - 90j$ for P, and $B = 1884$ for H1. The code forms this row by exact addition of the two source rows.

For every nonempty positive translate use the global quantiles

$$-v - l_v - M_p \le -90, \qquad v - u_v - S_p - M_q \le -89, \tag{6}$$

with reflected rows for negative centers. These are integer interval counts and require no assumption that a center lies in its target hull. For H1 ordinary quantiles are justified by $\delta \le 1$. Together with (1), (4), orders, and the inherited branch/contributor rows, (5) defines the coarse model. It has no K, no hole variable, no exposed inner points, and no minimum-linearization variables. We also omit the separate loss cap, joint span, and individual slot caps from every final model. Their scalar consequences still justify the ranges and refinements below.

3.1 A readable integer-row contradiction

The certificate mechanism can be seen without an optimizer. Consider Z0, branch A, with pair 2's positive contributor chosen. Write Dsi for the outer order row, Gs for $-R_s \le -r$, Hs for the doubled first row in (4), Lv for a lower quantile, C for (5), and L for (1). Take the following nonnegative row combination:

$$\begin{aligned} &2(\mathrm{Dp3} + \mathrm{Dq3}) + 6(\mathrm{Dp4} + \mathrm{Dq4}) + 2(\mathrm{Gp} + \mathrm{Gq}) + \mathrm{Hp} + \mathrm{Hq} \\ &\quad + 2(\mathrm{Lp4} + \mathrm{Lp9} + \mathrm{Lq5} + \mathrm{Lq8}) + 2\mathrm{C} + 2\mathrm{L}. \end{aligned}$$

Every resulting variable coefficient is nonnegative. The remaining outer coefficients are $2p_3 + 2q_3$; all other surviving terms are nonnegative loss coordinates. The right side is exactly

$$-2a^2 + 162a - 4072 = -792 - 2I < 0.$$

Thus this entire family is impossible. The other stored vectors are checked in exactly the same way, with their quadratic right sides maximized over their full integer parameter intervals.

4 Refine only when the coarse spine needs it

4.1 Root coverage and center conventions

Set $\lambda_p = p_1 - p_2 - p_3 - p_4$, $\alpha_p = p_2 - p_3 - p_4$, and $\beta_p = 2p_2 - p_1 - p_3 - p_4$. The positive center list is

$$v^p = (\lambda_p + p_1, \dots, \lambda_p + p_4, \lambda_p - q_1, \dots, \lambda_p - q_4, \alpha_p, \beta_p),$$

with the reflected list v^q. Labels `p0`,...,`q9` mean these centers, not the outer magnitude variables. Pair 1 is `p4,q4`. Branch A has $\beta_p + \beta_q \le 0$ and pair 2 is `p9,q9`; branch B has $\beta_p + \beta_q > 0$ and pair 2 is `p5,q5`. The other sixteen units are singles. The mode construction is the unchanged exhaustive

construction in [1]: enumerate the missing units, reflect a present pair 1 to its positive contributor, and enumerate pair 2's contributor. If branch A has a missing single `p5` or `q5`, a present pair 2 may split. Drop its individual loss rows, enumerate an actually contributing member s, and retain $-\beta_s - M_s \le -1$. This necessary row is not treated as a sufficient condition for contribution. There are respectively 4, 70, and 578 modes when $j = 0, 1, 2$.

Family	j	Integer a range	Root groups	Integer root models
Z0	0	26–55	4	120
P0	0	27–54	4	112
Z1	1	29–52	70	1680
P1	1	30–51	70	1540
Z2	2	34–47	578	8092
P2	2	36–45	578	5780
H1	1	40–41	70	140
Total			1374	17464

Every root must either have a certificate or undergo an explicitly allowed refinement. The certificate file cannot supply a smaller root universe: the checker reconstructs it and compares exact key sets.

4.2 Two loss rules, with no blanket large-center enumeration

A retained translate has $90 - \lambda_v$ hull points. If this number exceeds four, the core quantiles are valid:

$$-v - l_v - R_p \le -86, \qquad v - u_v - S_p - R_q \le -85.$$

The derivation is the core interval count in [1]: an outside translate using more than four hull points must use a core point, since only four points on that side lie outside the core.

When $j = 2$, the actual residual budgets are at most 47 in Z and 22 in P. Thus all centers use core quantiles. In P2 the stronger ordinary rows are valid because $\delta < \min(a, b) + 5$, the minimum outside loss:

$$-v - l_v \le -b - 5, \qquad v - u_v - S_p \le -a - 4.$$

This justifies the two-empty refinement, independently of whether the final linear model retains its loss-cap row.

When $j = 1$, the budgets are at most 137 in Z and 112 in P, both below $2 \cdot 86$. At most one retained center has $\lambda_v \ge 86$. The complete refinement is therefore: either every center uses core quantiles, or choose each retained center once as the unique large-loss center, impose $\lambda_v \ge 86$ there and use its global quantiles, and use core quantiles elsewhere. The large-center ranges are $a = 34, \dots, 47$ in Z and $a = 36, \dots, 45$ in P, as independently checked from the original scalar budgets. Deleting slot caps weakens these necessary models and causes no loss of coverage.

Only four roots actually require this split: Z1 and P1 in branch A with pair 2 missing, or with single `p0` missing and pair 2's negative contributor chosen. Each has 17 retained centers and produces 18 leaves. All 136 other one-empty low-hole roots are excluded directly by the global coarse model. In particular, the large-loss location is never enumerated for those roots.

4.3 The five leaves that need nested envelopes

In the delivered proof, only these leaves use more than the coarse sumset bound:

Family	Missing units	Pair-2 choice/member	Large center	a range
Z1	A: `p0`	negative contributor	`p8`	34–47
P1	A: `p0`	negative contributor	`p8`	36–45
Z2	A: `p0,p8`	negative contributor	none	34–47
Z2	A: `p0,q5`	negative actual member	none	34–47
P2	A: `p0,p8`	negative contributor	none	36–45

Expose $x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p$ and their negative analogues y_i, with integer gaps at least one. Use the two exposed mass rows

$$S_p \geq T_a + \varepsilon_p + \sum_{i=1}^{4} x_i, \qquad S_p \geq T_{a-1} + \sum_{i=0}^{4} x_i,$$

and their reflections. The two coarse mass rows follow from these and the first three gaps, so they need not be included again. The four gaps also imply $M_p \geq R_p + 4$. For $t = 1, 2, 3$ put $W_t = x_t + y_t$ and $s_t = 2(4 - t)$, and use

$$K \leq 2W_t + 1 + \binom{s_t + 1}{2} + \sum_{i=t+1}^{4} \left[\min(W_t + 1, x_i - x_{i-1}) + \min(W_t + 1, y_i - y_{i-1})\right]. \tag{7}$$

To prove this, replace the inner points through level t by their integer hull. Its double sum has $2W_t + 1$ points. Each successive external translate extends the union by at most the smaller of its gap and $W_t + 1$. At most $\binom{s_t+1}{2}$ pairs among the remaining external points remain. Introduce one nonnegative variable for each minimum, bounded above by both arguments. Actual sets choose the minima; equality is unnecessary in the relaxation. These leaves retain K, the coarse bound, and the original cardinality row. They use only three envelope levels; neither level 0 nor level 4 is needed. They introduce 19 additional coordinates: six inner points, K, and twelve minimum variables. No such coordinates occur in the other 1,437 leaves.

5 Exact verification and result

Every final model is $Ax \leq b(a)$, $x \geq 0$, with integer A and a rational quadratic right side. A stored nonnegative integer vector u is accepted only if $uA \geq 0$ and $\max_{a \in [L,U] \cap \mathbb{Z}} ub(a) < 0$. The maximum is computed at endpoints and, for a concave quadratic, the floor and ceiling of the vertex when in range. The polynomial is built symbolically; it is not fitted to sampled values.

Measure	Previous proof	This proof
Final structural leaves	3746	1442
Integer models covered	45928	18280
Interval certificate checks	3750	1455
Stored integer vectors	209	94
Leaves requiring nested coordinates	3746	5
Least normalized strict gap	12/349	4/85

The previous model imposed nested coordinates at every leaf; the table does not claim that every previous certificate used them. The new routing has 1,313 direct coarse roots, 54 two-empty coarse refinements, three two-empty nested refinements, and four loss splits. The splits have 70 coarse leaves and two nested leaves. The totals include every branch, not just successful representative instances. Model count falls by about 60.2%, interval checks by 61.2%, and stored vectors by 55.0%. These are proof-size comparisons, not timing claims.

The standard-library checker rejects a missing root, a missing actual contributor, a missing large-loss outcome, interval gaps or overlaps, invalid refinements, unknown rows, negative weights, and noncontradictory weights. It checks transport before eliminating the hole coordinate. The `-exhaustive` run additionally rebuilds every one of the 18,280 numerical models, compares every row with the symbolic formula, checks every row combination, and compares the minimum pointwise gap with the interval maximum. This run agrees with the interval result. Standard and `-O` result files are byte-identical; fresh certificate discovery reproduces the certificate file byte-for-byte; the previous upper-97 checker reproduces its saved result unchanged. Twelve negative controls pass. Actual commands, elapsed times, hashes, and comparisons are recorded in `results/2026-10-16/proof_method/`.

Theorem 5.1. *Assuming the audited foundation and structural reduction stated in [1], every finite real set A with $\mathrm{FS}(A) \subseteq 4A$ satisfies $|A| \leq 97$.*

Proof. The foundation excludes sizes above 98. A hypothetical size-98 set satisfies the unchanged scalar reduction. High-hole active cases are excluded there; the remaining high cases map to H1. Every low-hole case maps to one of the seven root families. Lemma 2.1 sends positive low holes to their endpoint models. A directly certified root is impossible. Every other root undergoes a valid exhaustive refinement by the two loss rules. The resulting necessary models, including all five nested exceptions, have strict exact integer-row contradictions. Therefore no size-98 set exists. □

This proof does not establish $N \leq 96$, and its verifier explicitly rejects a request to certify the next cardinality. The next research step is to apply the simplified coarse spine and isolate its genuinely necessary exceptions at size 97.

Reproduction. From the repository root, Python 3.10 or later suffices:

```
python -I -S -B verification/upper97_simplified.py
python -I -S -B verification/upper97_simplified.py --exhaustive
```

The optional generator requires SciPy; it is not imported by verification:

```
python experiments/generate_upper97_simplified_certificates.py
```

The Windows entry is `verify_upper97_simplified_windows.bat`. It runs the four preceding continuation checks and then this simplified size-98 exclusion. The earlier endpoint proof and its data remain available for comparison.

References

[1] Sum-of-Subset research project, *Subset Sums in a Fourfold Sumset: A unified loss and hole proof from 98 to 97*, 16 October 2026. Repository report, with `verification/upper97.py`; includes references to the audited 115–103–100–99–98 foundation and the archived strategy.